

Report for TP2: Converging Thin Optical Lenses

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1 Introduction

This TP aims to experimentally measure the focal length of a converging optical lens and explore its practical optical properties, notably in the formation of a virtual/real reflected image.

2 Experimental setup

For this TP, we have a graduated guide rail (1), three sliding lens mounts (2), a light source (3), a mirror (4), a gridded glass with sliding mount and detachable projection screen frame (5), a projection screen (6) and an additional converging lens (7) in addition to the converging optical lens (8) at the centre of our study.

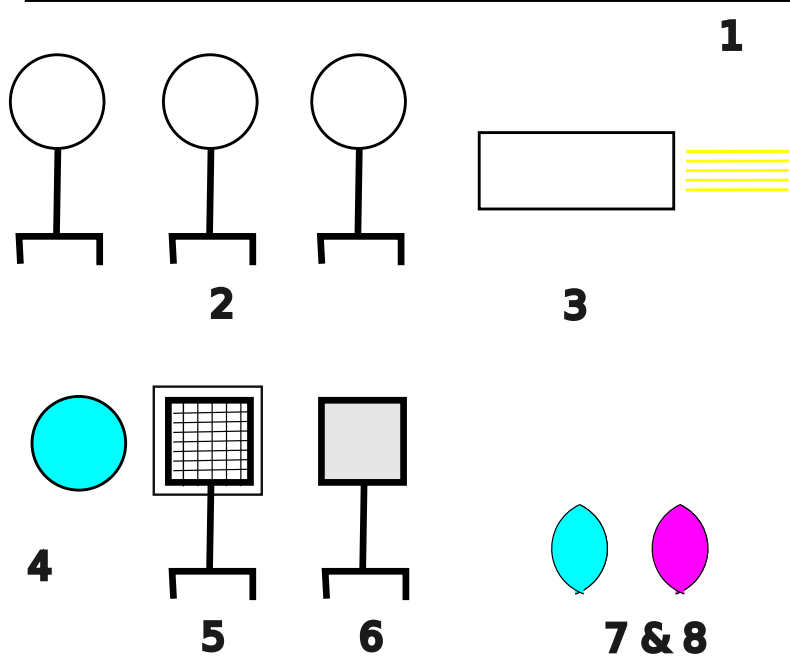


Figure 1: Guide rail and autocollimation setup.

3 Experimental procedure

3.1 Rough estimation (Autocollimation)

We begin the experiment with a rough estimation of focal length by placing the lens between a distant, patterned light source (room lighting with grid) and a screen (a white table). We have observed that the pattern enters into focus at around 200 mm measured by a meter ruler.

3.2 Refined estimation (Autocollimation)

We repeated the autocollimation method with our guided rail. The lamp placed behind the gridded glass serves as a light source, while the screen provides an identifiable pattern. The light is converged by the converging lens, reflected by the mirror, then reconverged once again to form the image on the attached screen on the grid. We adjust the grid until a clear image is obtained.

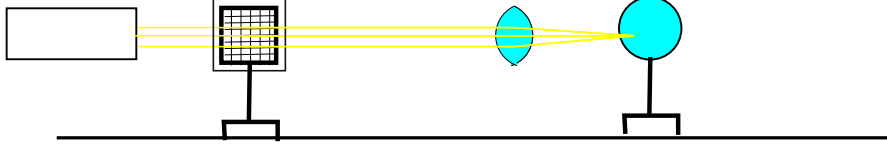


Figure 2: Refined autocollimation setup on guided rail.

We have thereon measured the focal length of the converging lens, which is equal to the distance between the grid and the lens, at 205 mm.

3.3 Measurement with improved precision

With Descartes's law:

$$\frac{1}{\overrightarrow{AO}} + \frac{1}{\overrightarrow{OA'}} = \frac{1}{\overrightarrow{OF'}} \Leftrightarrow \overrightarrow{OF'} = \frac{\overrightarrow{AO} \cdot \overrightarrow{OA'}}{\overrightarrow{AO} + \overrightarrow{OA'}},$$

we vary experimentally the distance of $\overrightarrow{AO}$ by moving the grid, identify $\overrightarrow{OA'}$ by moving the screen until a clear image is obtained, and calculate mathematically the focal length f which is equal to $\overrightarrow{OF'}$.

To maximise precision, we repeat the measurement with 10 different $\overrightarrow{AO}$.

3.3.1 Real image measurement

First, we set up the apparatus as such:

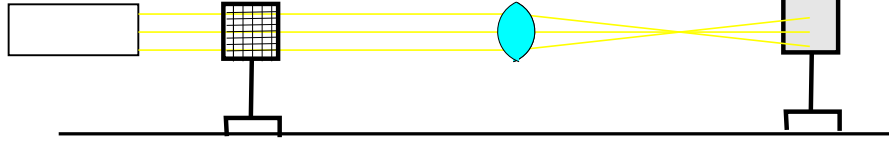


Figure 3: Setup for real image conjugation measurement.

So a real image is obtained on the screen.

Table 1: Real image measurement data (distances in mm).

$\overrightarrow{AO}$ (measured)	400	450	500	550	600	650	700	750	800	850
$\overrightarrow{AA'}$ (measured)	824	827	857	883	916	953	993	1033	1078	1125
$\overrightarrow{OA'} = \overrightarrow{AA'} - \overrightarrow{AO}$	424	377	357	333	316	303	293	283	278	275
$f = \overrightarrow{OF} = \frac{\overrightarrow{AO} \cdot \overrightarrow{OA'}}{\overrightarrow{AO} + \overrightarrow{OA'}}$	205.825	205.139	208.285	207.418	206.987	206.663	206.546	205.470	206.308	207.778

3.3.2 Virtual image measurement

As a virtual image cannot be directly visualised by a screen, we must use a second converging lens to perform indirect measurement.

Setting up as such, we adjust the object (grid) position (A) to an arbitrary value slightly smaller than the focal length (estimated to be 205 mm), then we adjust the second converging lens until a clear image of the grid is obtained on the screen.

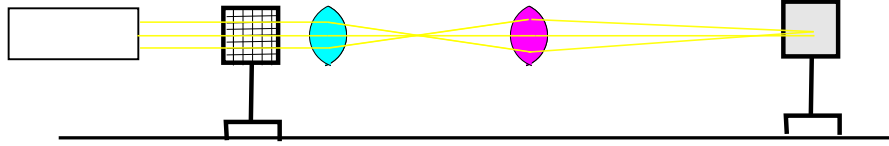


Figure 4: Setup for virtual image measurement (step 1: with both lenses).

Then we remove the converging lens to be measured (“1st lens”) and move the mirror until a clear image is obtained.

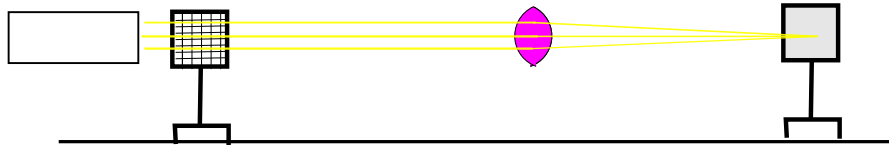


Figure 5: Setup for virtual image measurement (step 2: 1st lens removed).

The position of the mirror is the position of the virtual image A' .
We thereby obtain the following results, in mm:

Table 2: Virtual image measurement data (distances in mm).

A (ruler reading)	213	212	189	200	160	185	200	210	195	205
O (ruler reading)	300	300	300	300	300	300	300	300	300	300
A' (ruler reading)	146	140	46	110	-117	39	104	141	103	132
$\vec{AO} = O - A$	87	88	111	100	140	115	100	90	105	95
$\vec{OA'} = O - A'$	-154	-160	-254	-190	-417	-261	-196	-159	-197	-168
$f = \vec{OF} = \frac{\vec{OA} \cdot \vec{OA'}}{\vec{OA} + \vec{OA'}}$	199.970	195.556	197.161	211.111	210.758	205.582	204.167	207.391	224.837	218.630

4 Statistical analysis

4.1 Best estimate of true value

With 10 measurements of the same value, the true value is best estimated by the statistical mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Which we calculated to be 206.6 mm with the real image method, and 207.5 mm with the virtual image method.

4.2 95% confidence interval

The 95% confidence interval of the true value can be mathematically calculated as such:

$$\bar{x} \pm t_{95\%, df} \frac{s}{\sqrt{n}}$$

Where t is the 95% confidence value according to the Student's distribution for degree of freedom $n - 1 = 9$, s is the standard sample deviation calculated by

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}.$$

We have therefore the 95% confidence interval of the true value calculated as 206.7 ± 0.7 mm for real image measurement, and 207.5 ± 6.6 mm for virtual image measurement.

5 Results interpretation

The best estimates of the focal length obtained by the autocollimation, virtual image and real image conjugation methods concur. The real image method led to the best overall accuracy, while the autocollimation method provided a rapid estimate. The virtual image method is found to be less accurate, as it is practically difficult to manipulate the grid so it corresponds

exactly to the position of the virtual image, due to the fact that the image projected onto the screen is magnified and therefore difficult to discern whether a good focus is obtained.

We thereby decide that the value obtained by the real image conjugation method, 206.7 ± 0.7 mm, should be accepted as our best estimate of the true focal length.

6 Conclusion

In this experiment, with the aid of a graduated guide rail and various optical equipment, we indirectly measured the focal length of a converging lens by three methods: autocollimation, real image conjugation, and virtual image conjugation. Our best estimate of the focal length f is 206.7 mm.

Bibliography